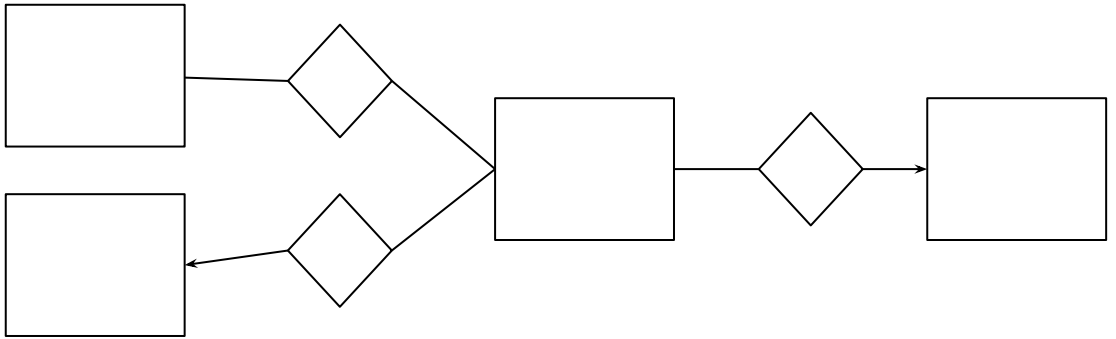
CSC 370: Databases Weak Entity Sets & Inheritance

Sean Chester • University of Victoria

How many relations are implied?



❑ 0 ❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5

Today's Goal:

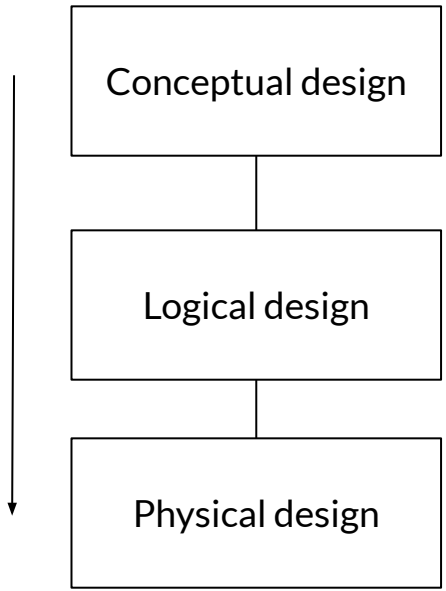
You can simplify conceptual designs using entity sets that are defined in terms of other entity sets.

Outline

1. Review ERD quickly
2. Inheritance in conceptual design
3. Weak entity sets
4. Design Exercises (first in pairs, then together)

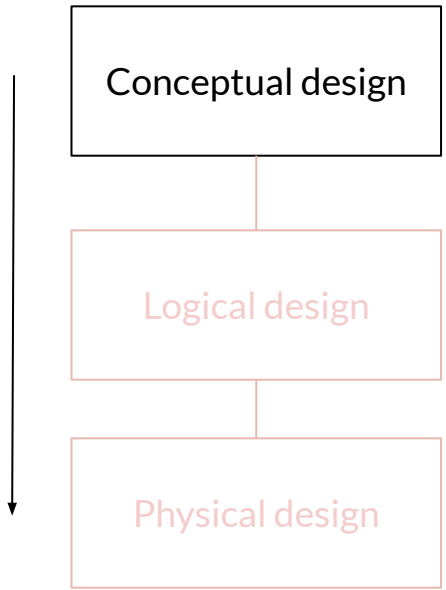
Review:
ERD Basics

Conceptual Design



- Defining what should be in the database
- Defining how it maps onto the relational (or other) data model
- Implementing the logical schema concretely, e.g., data types

Conceptual Design



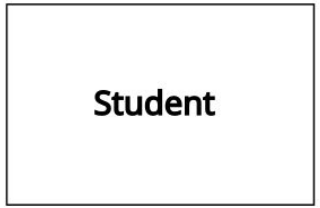
- Defining what should be in the database
  - Independent of choice of logical model
  - Accessible to many stakeholders
  - Documentation that gives broad understanding of system at a quicker glance

ERD's: entity sets

An ERD is a *conceptual model* that only describes broadly the entities in a database and their relationships.

An *entity set*, analogous to a "class" in OOP, is "an abstract object of some sort" (pg. 126) and denoted by a rectangle with the name of the entity set inscribed therein.

The following *very simple* ERD has one entity set representing *students*.



Source: Garcia-Molina et al. (2009)

ERD's: attributes

An ERD is a *conceptual model* that only describes broadly the entities in a database and their relationships.

An *attribute*, somewhat analogous to a "member variable" in OOP, are properties of the entity set to which they are affiliated. They are denoted by an ellipse with the name of the attribute inscribed therein and a line connecting it to the relevant entity set.

The following *very simple* ERD has one entity set representing *students* and two attributes: v_number and name.



Source: Garcia-Molina et al. (2009)

Identifiers

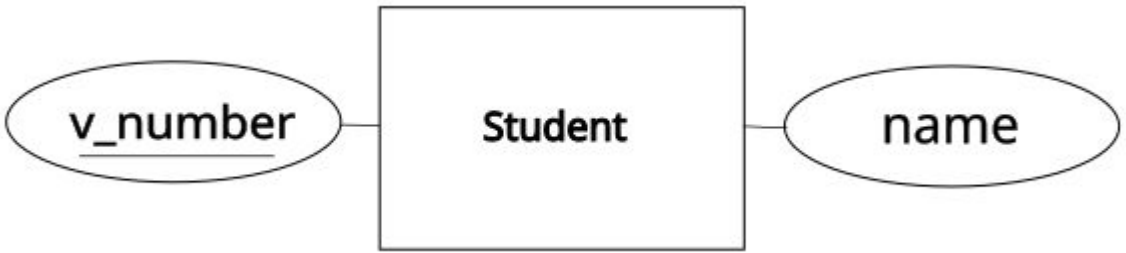
A set of attributes A is an *identifier* of entity set E iff:

- There cannot be two instances of E that have the same value for all attributes in A
- A is *minimal* (i.e., no subset of A meets the first condition)

We should assign an identifier to every entity set.
we underline all attributes that are part of the identifier

- Clearly, then, it is not possible to have more than one identifier

Here, it is not possible for two Students to have the same student number (v_number). Thus {v_number} is a "simple" (i.e., single-attribute) identifier.



Source: Garcia-Molina et al. (2009), §4.3.2

Identifiers

A set of attributes A is an *identifier* of entity set E iff:

- There cannot be two instances of E that have the same value for all attributes in A
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- Clearly, then, it is not possible to have more than one identifier

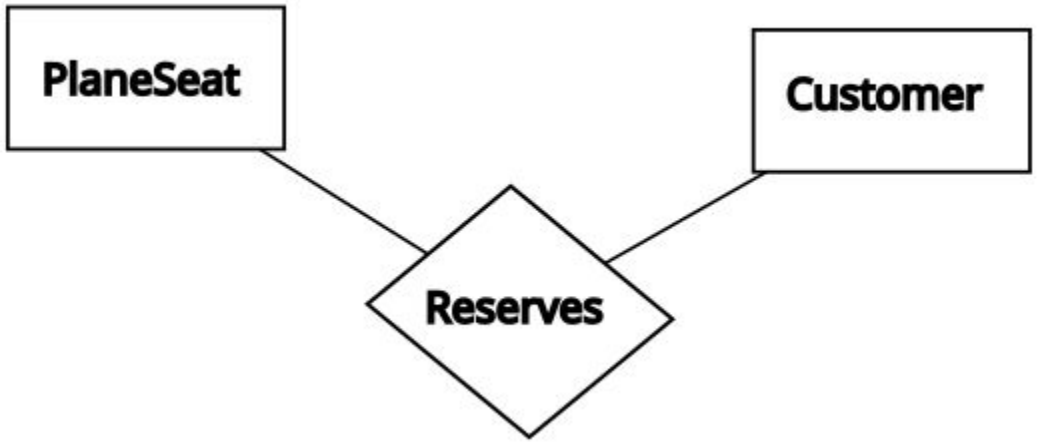
Here, any two students could have the same name, but they cannot also have the same birthdate. This identifier is composite, because the identifier has many attributes.



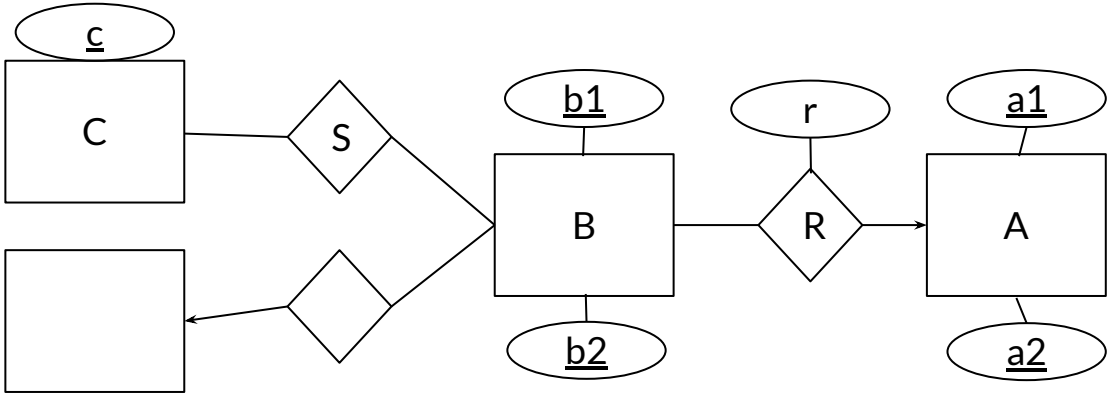
Source: Garcia-Molina et al. (2009), §4.3.2

Constraints on relationships

- *Here's a more serious example. What's wrong with this ERD?*
  - *Without modelling constraints, there is no limit on how many customers can reserve the same seat*
    - *(Better get to the airport early!)*



ERDs to SQL



With annotation tool, write SQL for A, B, C, R, and S

Generalisation

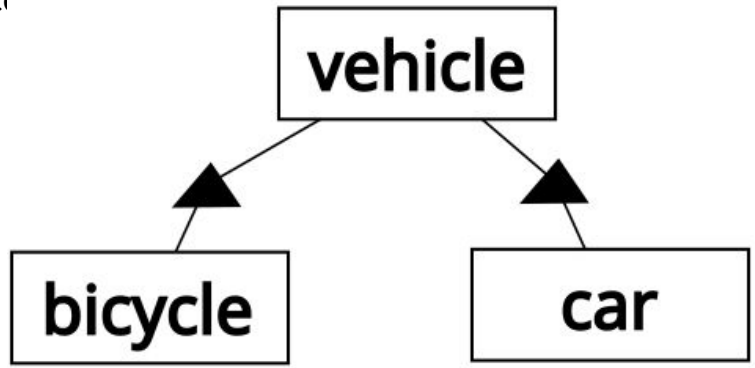
Generalisation

Generalisation defines one concept as being a more specific type of another concept (which you've seen as *inheritance* in object-oriented programming)

- The concept *vehicle* is more general than the concepts of *bicycle* and *car* or a *electric* and *combustion*.
- The concept of an *employee* is more general than the concepts of *manager*, *clerk*, and *professor*

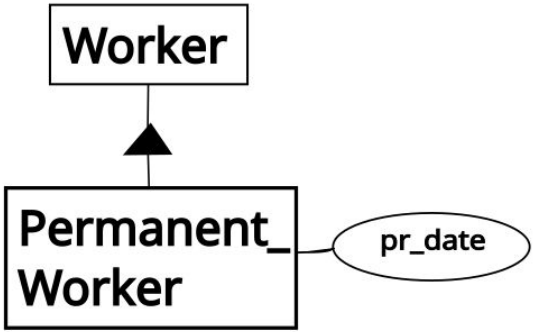
Importantly, with generalisation, all properties of the *parent* concept are inherited by all of the children

We represent subsets with a black arrow pointing to the parent concept



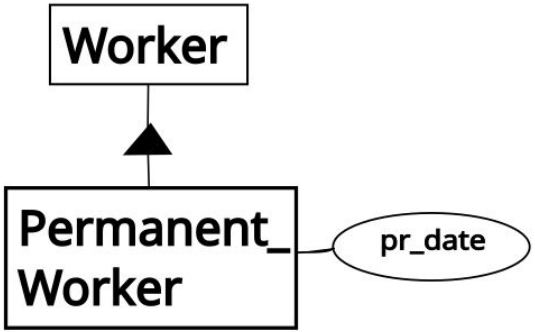
Why use subsets?

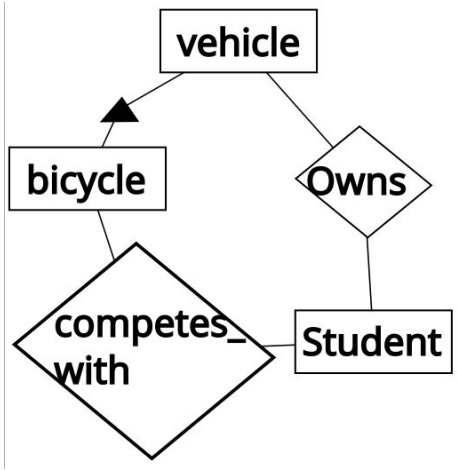
- This arises specifically when there is a subset of an entity set to which additional attributes or an additional relationship applies
  - In the ERD below, we see that for permanent (visa) workers, we also want to track the date they received permanent residence (pr_date), which is not relevant to other workers



Why use subsets?

- This arises specifically when there is a subset of an entity set to which additional attributes or an additional relationship applies
  - In the ERD below, we see that for permanent (visa) workers, we also want to track the date they received permanent residence (pr_date), which is not relevant to other workers
  - In the ERD to the right, creating a subset allows us to construct a more expressive relationship that can only involve certain types of vehicles



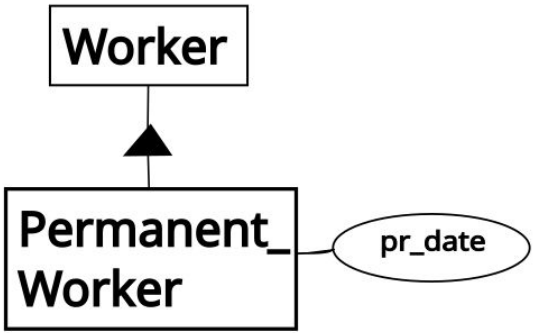


Subsets as tables

- A subset is easy to recognise in SQL DDL because the primary key is also a foreign key

```
CREATE TABLE `Worker`(`id` int PRIMARY KEY);

CREATE TABLE `PermanentWorker`(
   `id` int PRIMARY KEY,
   `pr_date` DATETIME NOT NULL,
  FOREIGN KEY(`id`) REFERENCES `Worker`(`id`)
);
```

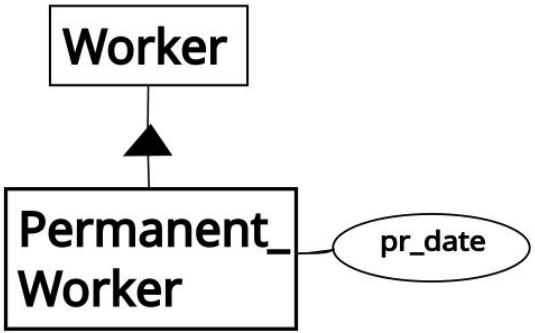


Subsets as tables

- A subset is easy to recognise in SQL DDL because the primary key is also a foreign key

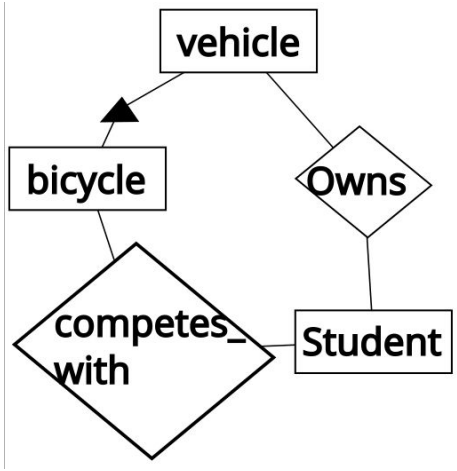
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CREATE TABLE `Worker`(`id` int PRIMARY KEY);

CREATE TABLE `PermanentWorker`(
  `id` int PRIMARY KEY,
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  FOREIGN KEY(`id`) REFERENCES `Worker`(`id`)
);
```



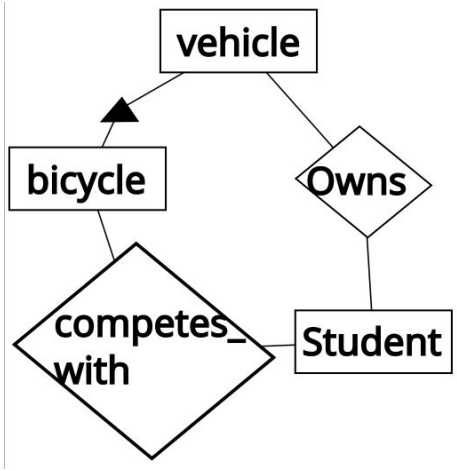
Subsets as tables

- What is the SQL DDL to implement this ERD?



Subsets as tables

- What is the SQL DDL to implement this ERD?




```
CREATE TABLE `Vehicle`(`id` int PRIMARY KEY);
CREATE TABLE `Student`(`id` int PRIMARY KEY);

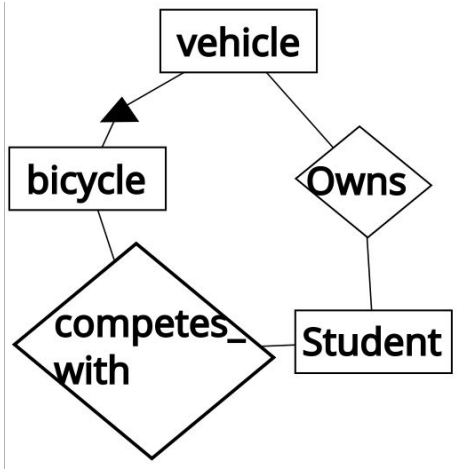
CREATE TABLE `Bicycle`(
  `id` int PRIMARY KEY,
  FOREIGN KEY(`id`) REFERENCES `Vehicle`(`id`)
);

CREATE TABLE `CompetesWith`(
  `bicycle` int, student int,
  FOREIGN KEY(`bicycle`) REFERENCES `Bicycle`(`id`),
  FOREIGN KEY(`student`) REFERENCES `Student`(`id`),
);

CREATE TABLE `Owns`(
  `vehicle` int, student int,
  FOREIGN KEY(`vehicle`) REFERENCES `Vehicle`(`id`),
  FOREIGN KEY(`student`) REFERENCES `Student`(`id`),
);
```

Subsets as tables

- What is the SQL DDL to implement this ERD?



These are the same values, but
From different tables for ref. integrity!

```
CREATE TABLE `Vehicle`(`id` int PRIMARY KEY);
CREATE TABLE `Student`(`id` int PRIMARY KEY);

CREATE TABLE `Bicycle`(
  `id` int PRIMARY KEY,
  FOREIGN KEY(`id`) REFERENCES `Vehicle`(`id`)
);

CREATE TABLE `CompetesWith`(
  `bicycle` int, student int,
  FOREIGN KEY(`bicycle`) REFERENCES `Bicycle`(`id`),
  FOREIGN KEY(`student`) REFERENCES `Student`(`id`),
);

CREATE TABLE `Owns`(
  `vehicle` int, student int,
  FOREIGN KEY(`vehicle`) REFERENCES `Vehicle`(`id`),
  FOREIGN KEY(`student`) REFERENCES `Student`(`id`),
);
```

Weak Entity Sets

Weak Entity Sets

- A *strong* entity set E, like those on the previous slide, is one in which the identifiers are a subset of the attributes of E
- A *weak* entity set E, instead, has as part of its identifier attributes that come from some other entity set E'
- A weak entity set borrows the entire key of some entity set to which it is related. Thus, we build a composite attribute that connects to the relationship line

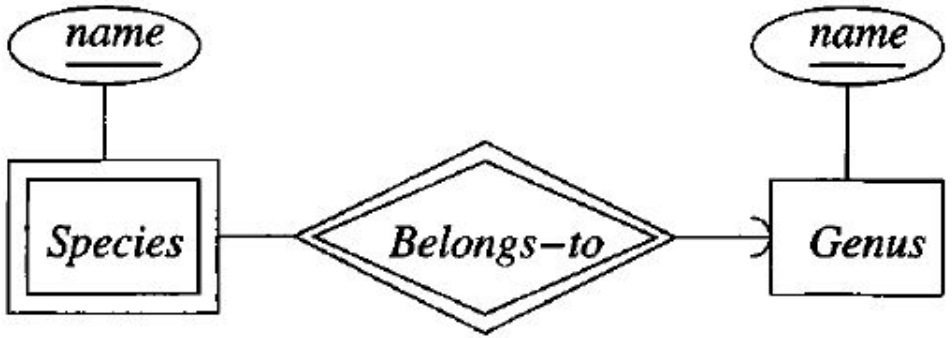


Figure 4.21: Another weak entity set, for species

Source: Garcia-Molina et al. (2009), §4.4

Weak Entity Sets

- A *strong* entity set E, like those on the previous slide, is one in which the identifiers are a subset of the attributes of E
- A *weak* entity set E, instead, has as part of its identifier attributes that come from some other entity set E'
- We illustrate a weak entity set with a double rectangle and a double diamond on the relevant relationship

What are the identifiers in the schema given below?

- Genus has a single rectangle. Therefore it is a strong entity set with {*name*} as the identifier
- Species has a double rectangle and is *supported by* relationship *Belongs-to*. Therefore it is a weak entity set with {*name*, *Genus.name*} as its identifier.

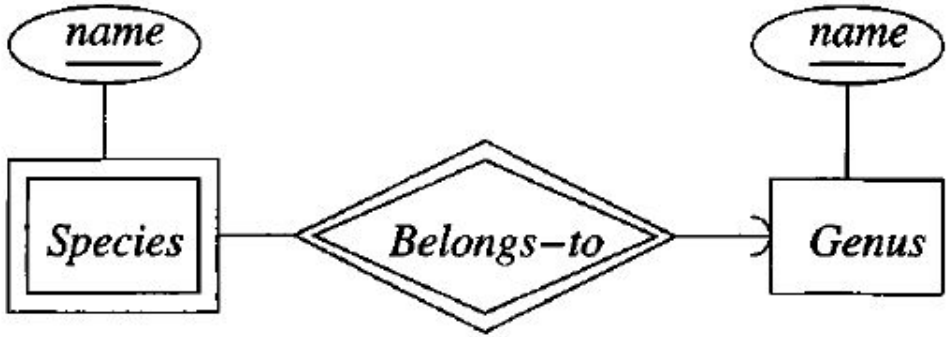


Figure 4.21: Another weak entity set, for species

Source: Garcia-Molina et al. (2009), §4.4

Weak entity sets as tables

- A weak entity set is easy to recognise in SQL DDL because *part of* the primary key is also a foreign key

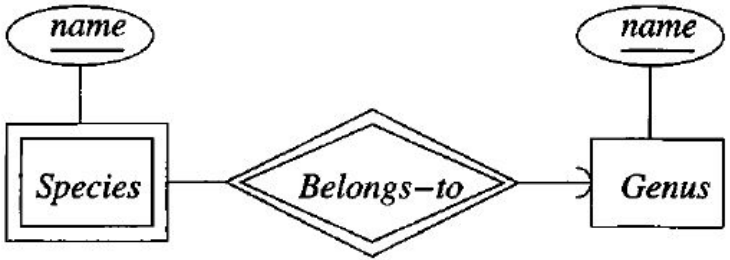


Figure 4.21: Another weak entity set, for species

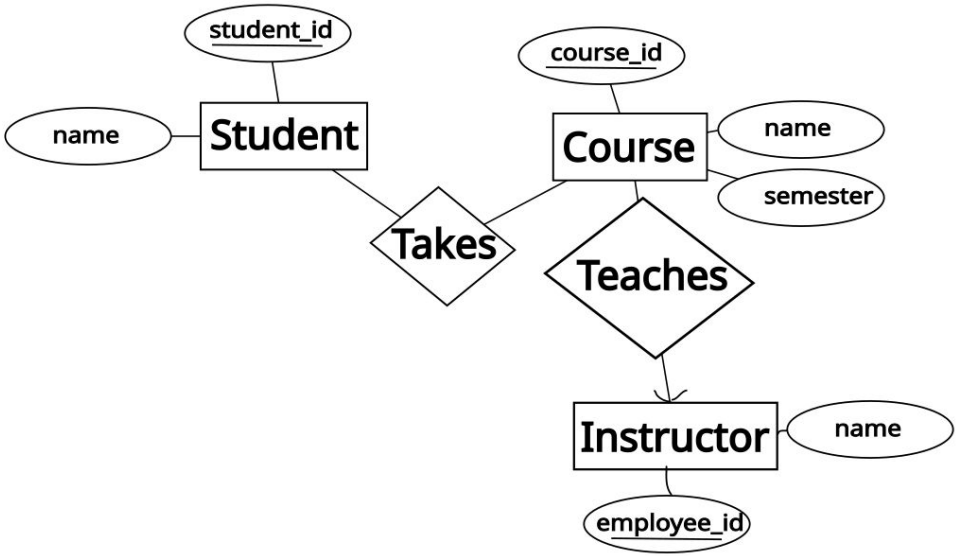
```
CREATE TABLE `Genus`(
  `name` varchar(60) PRIMARY KEY
);

CREATE TABLE `Species`(
  `genus` varchar(60),
  `name` varchar(60),
  PRIMARY KEY(`genus`, `name`),
  FOREIGN KEY(`genus`)
    REFERENCES `Genus`(`name`)
);
```

Group Design Discussion

Pair Activity 1

How could we record which students are SENG?
What is the SQL DDL code corresponding to that?

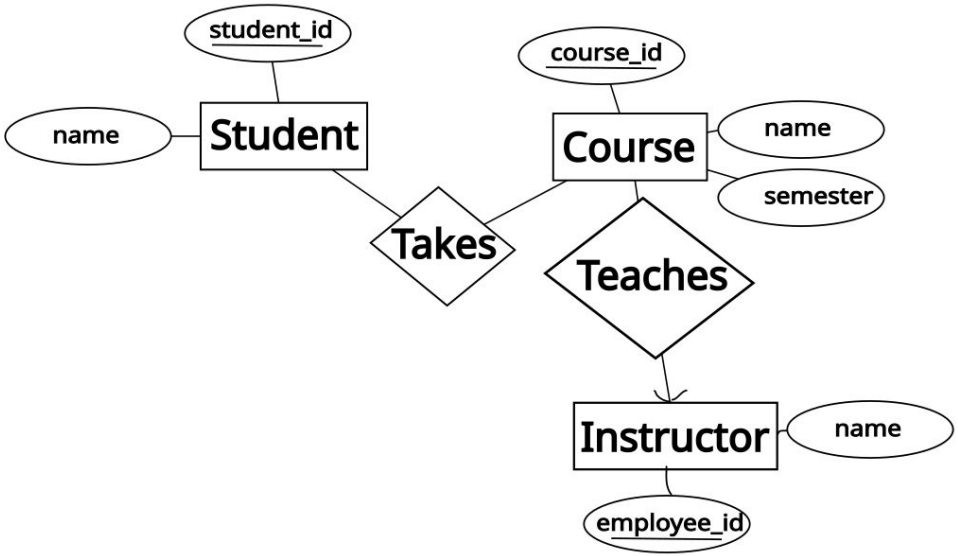


Allotted time: 180 seconds

Discussion 1

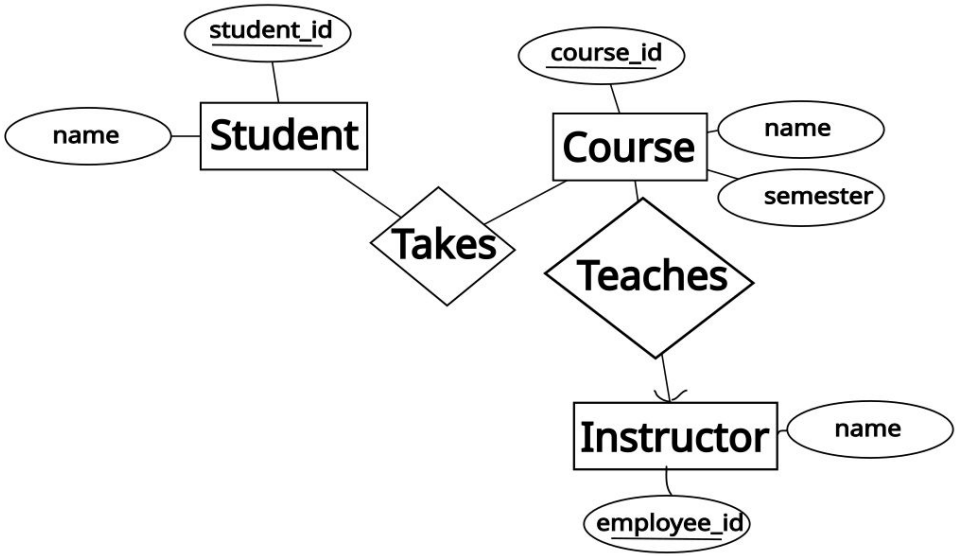
How could we record which students are SENG?

- Is a subset consistent with how this information is used?
- What if we want to add other majors?
- How do we restrict enrolment to SENG students?



Pair Activity 2

How should we extend this for an honour's thesis course?
What is the SQL DDL code corresponding to that?

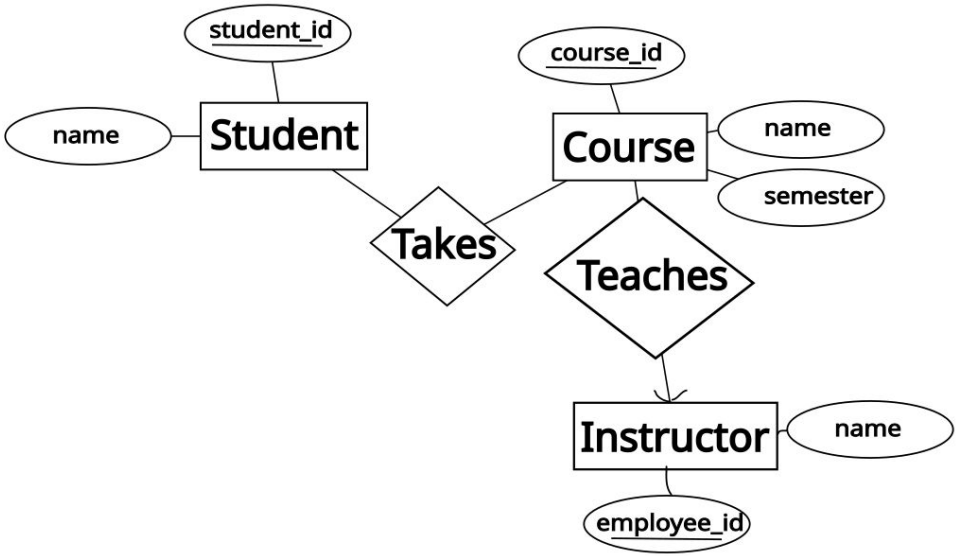


Allotted time: 180 seconds

Discussion 2

How should we extend this for an honour's thesis course?

- We need to add thesis_title, thesis_supervisor
- We need to ensure a student can only write one honour's thesis
- We want to know which students are honour's students, even if they have not yet registered for CSC 499

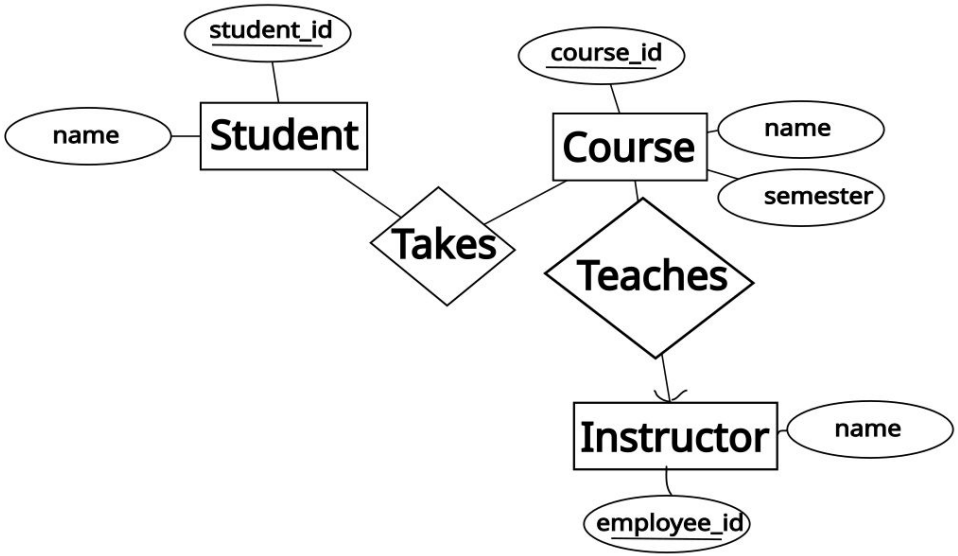


What is wrong with is_honours, thesis_title and thesis_supervisor as attributes of student?

- (hint: how are these related to each other?)

Pair Activity 3

How could we record which department (e.g., CSC) offers a course?
What is the SQL DDL code corresponding to that?



Allotted time: 180 seconds

Discussion 3

How could we record which department (e.g., CSC) offers a course?

- Does every course belong to a department?
- How will we handle interdisciplinary courses?

